MARK SCHEME · TEACHER COPY

10BR · Chapter 4: Software

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

Mark scheme extracts © Cambridge University Press & Assessment.

Total: 75 marks

Question 1 · May/June 2023 Paper 11, original Q4 [4 marks]

Question	Answer	Marks
1	**One** mark for each correct term in the correct place: • System • Application • Operating • Hardware	4

Question 2 · May/June 2023 Paper 12, original Q10 (a),(b) [5 marks]

Question	Answer	Marks
2(a)	**Two** from: • System software provides services that the computer requires • … whereas application software provides services that the user requires **One** from (system software): • Utility software // by example e.g. defragmentation software, antivirus, firewall • Operating system **One** from (application software): • Any suitable example of an application e.g. word processor, web browser, video-editing software	4
2(b)	• Secondary storage // HDD // SSD	1

Question 3 · May/June 2023 Paper 13, original Q4 (a),(b) [5 marks]

Question	Answer	Marks
3(a)	• B	1
3(b)(i)	• Machine code // low-level language // object code	1
3(b)(ii)	• Interpreter	1
3(b)(iii)	• Compiler	1
3(b)(iv)	• Compiler	1

Question 4 · Oct/Nov 2023 Paper 11, original Q9 (a),(b),(c) [10 marks]

Question	Answer	Marks
4(a)	Any **one** from: – Operating system // Interrupt handler	1
4(b)	Any **five** from: e.g. – Key press generates the interrupt – Interrupt given a priority – Interrupt is sent to CPU – Interrupt is placed in a queue – CPU stops current task to check the queue/service the interrupt … – … using an interrupt service routine – If key press is highest priority the interrupt is processed	5
4(c)(i)	Any **two** suitable hardware example e.g.: – Moving the mouse – Clicking a mouse button – Plugging in a device – Paper jam in printer – Printer out of paper	2
4(c)(ii)	Any **two** suitable software examples e.g.: – Division by zero – Two processes accessing the same memory location – Null value	2

Question 5 · May/June 2024 Paper 12, original Q4 (a),(b),(c) [5 marks]

Question	Answer	Marks
5(a)	Any **one** from: • Operating system • Utility software // by example	1

Total: 75 marks

5(b)	Any **two** from.: Example: • Word processor • Spreadsheet • Database • Web browser • Image/photo-editor • Video-editor • Email client • Game • Social media app	**2**
5(c)	Any **two** from: • System software **manages/maintains** the **hardware/software** • Applications software allows the **user** to perform tasks	**2**

Question 6 · Oct/Nov 2024 Paper 11, original Q7 (a),(b) [4 marks]

Question	Answer	Marks
6(a)(i) i)	D	1

Total: 75 marks

6(a)(ii)	**One** mark for identifying a function. **One** mark for a matching description. For example: • Memory management • … managing what gets allocated where in memory • Managing peripherals and drivers • … managing the communication between any input and output devices that are connected to the computer • Multitasking • … managing the process of **switching** between tasks that are being carrying out • Platform for running applications • … allowing communication between the applications software and the hardware • System security • … proving features such as username and password • User accounts • … allowing multiple user accounts to be created on a computer	**2**
6(b)	Any **one** from: • Firmware • Bootloader • BIOS • Bootstrap	**1**

Question 7 · Oct/Nov 2024 Paper 12, original Q1 (a),(b),(c) [3 marks]

Question	Answer	Marks
7(a) i)	**Instructions/program** that is used to operate a computer/hardware	1
7(b)	B	1
7(c)	Operating system // system software	1

Question 8 · Feb/March 2025 Paper 12, original Q7 (a),(b),(c) [9 marks]

Question	Answer	Marks
8(a)	• Language: assembly • Translator: assembler	2
8(b)	Any four from • Each statement is translated and executed before the next • **Translator stops** when an error is found • … can correct and continue from where error found // **real time** debugging when/where error found • ... does not need to retranslate all the code again • Can test sections of the code • Can test without all the program code being functional/accurate/complete	4
8(c)	Any three from: Example • Code editor • Run-time environment • Error diagnostics // by example • Auto-complete • Auto-correct • Prettyprint // syntax highlighting • Collapse/expand block • Auto-**documentation**	3

Question 9 · May/June 2025 Paper 13, original Q1 (b) [1 marks]

Question	Answer	Marks
9(a) i	A	1

Question 10 · May/June 2025 Paper 13, original Q4 (a),(b),(c) [6 marks]

Question	Answer	Marks
10(a)	Any **two** from: • It assigns a priority to an interrupt. • It uses an interrupt handler / interrupt service routine to process the interrupt. • Maintains/organises the interrupt queue	**2**
10(b)	Any **two** from: Examples: • managing files • providing an interface • managing peripherals and drivers • managing memory • managing multitasking • providing a platform for running applications • providing system security • managing user accounts	**2**
10(c)(i)	Any **one** from: Examples: • key press on a keyboard • button clicks on a mouse • printer out of ink	**1**
10(c)(ii)	Any **one** from: Examples: • division by zero • two processes trying to access the same memory location	**1**

Total: 75 marks

Question 11 · Oct/Nov 2025 Paper 12, original Q3 (a),(b),(c) [4 marks]

Question	Answer	Marks
11(a)	Any **two** from: • Low-level language • Uses mnemonics • Used to **communicate directly** with the computer hardware	2
11(b)	Assembler	1
11(c)	B	1

Question 12 · May/June 2026 Paper 11, original Q5 (a),(b),(c),(d) [10 marks]

Question	Answer	Marks	Guidance
12	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		
12(a)	To translate/convert high-level language into low-level language/machine code/binary	1	Just translates code/program to low-level/machine code is NE
12(b)	Assembler	1	Mark the first answer given. Do not accept 'assembly'.
12(c)	**One** mark for each correct letter **One** mark for each corrected statement • A • An interpreter translates and executes the code **one line/statement at a time** • E • A compiler produces an **executable** file	4	If an incorrect letter is given, do not read on to statement correction. Accept minor variations for correct statements that have the same meaning. For letter A we are looking for a correction on the use of variable. For letter E we are looking for a correction on the use of non-executable file. If no answer is given in answer space, look to see if indication is given in the table.

12(d)	**One** mark for each correct function **One** mark for a matching purpose • Code editor • Allows the **user/programmer** to write/change the program • Run-time environment • Allows the user to **run/execute** the code and see the **output/result** // Allows the user to **input data** to **run/execute** the code // To allow the user to test the code • Error diagnostics // debugging tools // by example e.g. break points, stepping, variable watch window • Finds/identifies/detects errors in the code // by example • Auto-completion • Gives **options** to complete the remainder of a command word/key term/identifier/syntax (to limit errors/speed up development) // It **predicts** the rest of a command word/key term/identifier (to limit errors/speed up development) // It **predicts** the required syntax e.g. when a bracket is needed (to limit errors/speed up development) • Auto-correction • Corrects **spelling** errors in command words/key terms/identifiers/syntax (to limit errors in the code) • Prettyprint // syntax highlighting // colour coding • Changes the colour of text in the code to highlight command words/key terms/identifiers (to make them easier to identify) • Line numbers • To give (incremented) numbers to each line in the code (to allow a line in the code to be easily/quickly found)	**4**	Do not accept translator/interpreter/compiler, this is given in the question. Mark first function given. If the function name is TV/blank, read on to the purpose to see if a correct function name is included. Award both correct name and purpose if seen. Award the purpose if the function name is still TV but it is obvious what is meant. Code editor – BOD just editor alone Code editor purpose – BOD edit in place of write/change Error diagnostics – just 'debugging' alone is TV Just 'words' alone is TV for command word/key term/identifier/syntax Allow minor spelling variations on function names. It you see something that you think is a viable alternate to the function name, please refer to your TL. Other viable functions outside the syllabus may be seen, such as: • artificial intelligence • auto-documentation • expand/collapse code blocks • built-in libraries/functions In this case, accept the function and a correct description of the purpose. The description must be specific and not generic. Refer to TL if in doubt.

Question 13 · May/June 2026 Paper 12, original Q7 (a),(b),(c),(d),(e) [9 marks]

Question	Answer	Marks	Guidance

13(a)	A language that is closer to the language processed/used by a computer
	// A language that is closer to binary/machine code
	// A language has that little abstraction from that processed by a computer
		1	Do not accept 'understood' in place of processed/used.
			
			'it uses/is binary' is TV for it is closer to binary.
			
			Do not accept that it is a language that uses mnemonics. This is not answering this question as not all do.
			
			Do not accept benefits e.g. it can directly manipulate the hardware. This is not answering the question.
			
			Do not accept marks for saying what it is not e.g. It is not close to human language.

13(b)	**Two** from: • Assembly (language) • Machine code // Binary • Object code	2	Mark first answer given. MP2 – BOD machine language
13(c)	Assembler	1	Mark first answer given.
13(d)	**Two** from: • The program doesn't need to be portable • Can **directly manipulate/access** the hardware // Can have low-level **control** of the device • The program will be faster to **execute/run** • The program can be written for **specific** hardware // Can use/access **specialised** hardware • The program uses **memory** more **efficiently** // The program is less memory intensive • The program would take up less **storage** space (in the vending machine) // The **file size** would be smaller (than the high-level equivalent file)	2	Mark whole answer. MP2 – program can communicate with hardware is TV as high-level language can also do this. There is no reference to it being direct. MP3 – Less time to translate is NE for execute MP5 – uses less space in memory is TV
13(e)	• Auto-correction • Run-time environment • PrettyPrint	3	Mark first answer given.